Ryon Marzbani

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Computer Science student at UBC experienced in building full-stack web applications, REST APIs, and AI-driven tools.

Experience

Software Engineering Apprentice Jun. 2026 – Sep. 2026
DataRealm *Vancouver, BC*

- Completed DataRealm's software engineering onboarding program, training on production testing standards, CI/CD automation, full-stack architecture, and agentic AI workflows.
- Built REST API backend services using Node.js, Express, and TypeScript following a layered architecture (routes, services, repositories) with typed PostgreSQL queries.
- Engineered automated test suites using Vitest and Jest while configuring GitHub Actions CI/CD pipelines for linting, type-checking, testing, and artifact deployment.

Education

University of British Columbia (UBC) Vancouver, BC
Bachelor of Science in Computer Science *Sep. 2024 – May 2028 (Expected)*

- **Relevant Coursework:** Data Structures and Algorithms, Software Construction, Computer Systems, Discrete Mathematics

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C++, SQL, HTML/CSS, Dart, Kotlin, Swift
Frameworks & Libraries: React, React Native, Flutter, Node.js, Express, Jest, Scikit-learn, NumPy, Pandas
Tools & Technologies: Linux, Docker, Kubernetes, PostgreSQL, Redis, Firebase, Cloud Infrastructure, REST APIs, Git
Concepts: Frontend & Backend Architecture, AI, Machine Learning, Natural Language Processing (NLP), Information Security, End-to-End Systems, Data Structures & Algorithms, Distributed Systems, Multi-Threaded Systems, OOP, WCAG Accessibility

Projects

AnswerOS — AI Search Visibility Platform | getanswerOS.com | GitHub | *React, TypeScript, Node.js, Redis, Stripe*

- Engineered a multi-model AEO platform used by **60+ users** to track brand recommendations across 6 LLMs (ChatGPT, Claude, Gemini, Groq, NVIDIA NIM, and Perplexity).
- Architected an async scanning pipeline using **Trigger.dev background tasks** and **Upstash Redis**, computing server-side 0–100 visibility scores from mention rate, rank, and sentiment.
- Implemented **Clerk authentication**, **Stripe billing** webhooks, evidence-based recommendation engines, and telemetry analytics.

Full-Stack Task Manager Monorepo | GitHub | *React, TypeScript, Vite, Node.js, Express, PostgreSQL, Claude API*

- Architected an npm monorepo task manager with a layered backend (routes, services, repos) and shared TypeScript types between Express API and React frontend.
- Engineered CRUD REST endpoints in Node.js using typed PostgreSQL queries via `pg` and Zod validation, enforcing $\geq$70% test coverage across Vitest and Playwright.

Vancouver Transit Companion App | GitHub | *React Native, TypeScript, GTFS-RT, Node.js, Express*

- Built a real-time transit tracking app covering 240+ routes using React Native, TypeScript, and GTFS-RT protobuf feeds.
- Implemented GPS-based ETA calculations using the Haversine formula and dynamic map visualization for live vehicle tracking.

Snap-Solve — Campus Connect App | GitHub | *React Native, TypeScript, Firebase, Cloud Functions*

- Built a React Native student networking app implementing geohash-based proximity matching for location-aware study group discovery.
- Designed a real-time request and messaging system using Firestore listeners and Firebase Cloud Functions to handle asynchronous event triggers.
- Engineered server-side validation and auto-expiring session logic within Cloud Functions, securing user data privacy and reducing database write overhead.